**Genome Explortion II**
Basic Genome Structure and Sequence Exploration Using Galaxy
Species: Callithrix jacchus (Common Marmoset)

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## Part 1. Reconfirm the Genome Identity

| **Field** | Value |
|---|---|
| **Species name** | *Callithrix jacchus* (Common Marmoset) |
| **NCBI Assembly accession** | GCF_049354715.1 |
| **Assembly level** | Chromosome |
| **FASTA filename in Galaxy** | Callithrix_jacchus.fasta |
| **Genome source/database** | NCBI (RefSeq) |
| **Approximate file size** | ~2.9 GB |

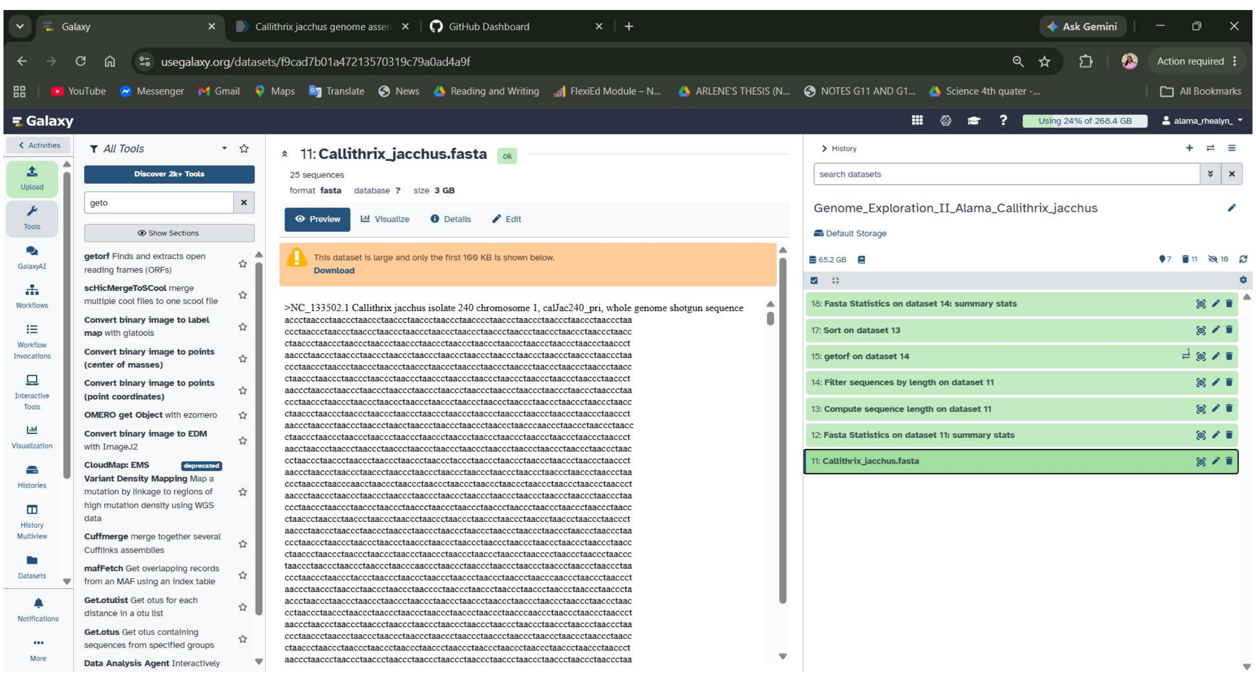

**Figure 1.** Genome dataset sitting on galaxy history.

**Part 2. Basic Assembly Statistics**

| **Metric** | Value |
|---|---|
| **Total assembly length** | 2,933,425,964 bp (~2.9GB) |
| **Number of sequences/contigs/scaffolds** | 25 |
| **Minimum sequence length** | 16,499 bp |
| **Maximum sequence length** | 220,781,134 bp |
| **Mean/average sequence length** | 117,337,038 bp |
| **N50** | 141,933,726 bp |
| **L50** | 9 |
| **GC content (%)** | 40.83% |

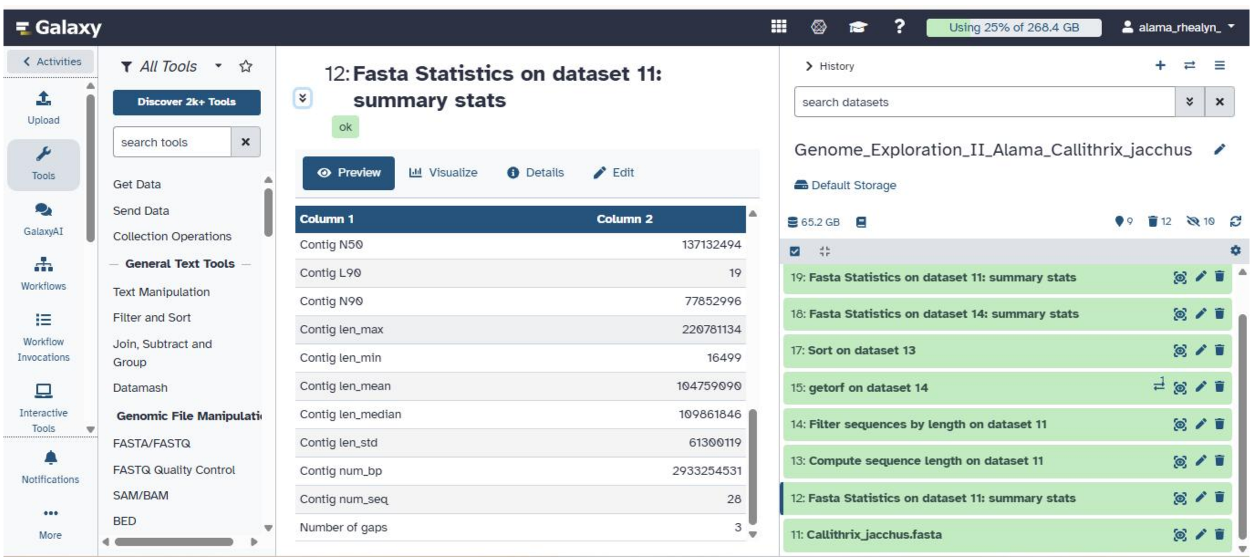


**Figure 2**. Statistics output

### Part 3. Sequence-Length Structure

| Rank | Sequence ID | Length (bp) |
|---|---|---|
| 1 | NC_133502.1 Callithrix jacchus isolate 240 chromosome 1, calJac240_pri | 220,781,13 |
| 2 | NC_133503.1 Callithrix jacchus isolate 240 chromosome 2, calJac240_pri | 206,717,255 |
| 3 | NC_133504.1 Callithrix jacchus isolate 240 chromosome 3, calJac240_pri | 193,242,953 |
| 4 | NC_133505.1 Callithrix jacchus isolate 240 chromosome 4, calJac240_pri | 178,191,855 |
| 5 | NC_133506.1 Callithrix jacchus isolate 240 chromosome 5, calJac240_pri | 167,432,195 |

The genome is mostly represented by a few very long sequences rather than many short ones, just 9 of the 25 scaffolds make up half the total genome (L50 = 9, N50 = 141.9 Mb), and the top 5 sequences alone are 167–220 Mb long, consistent with full chromosomes, which is a pattern that is typical for a chromosome-level assembly.

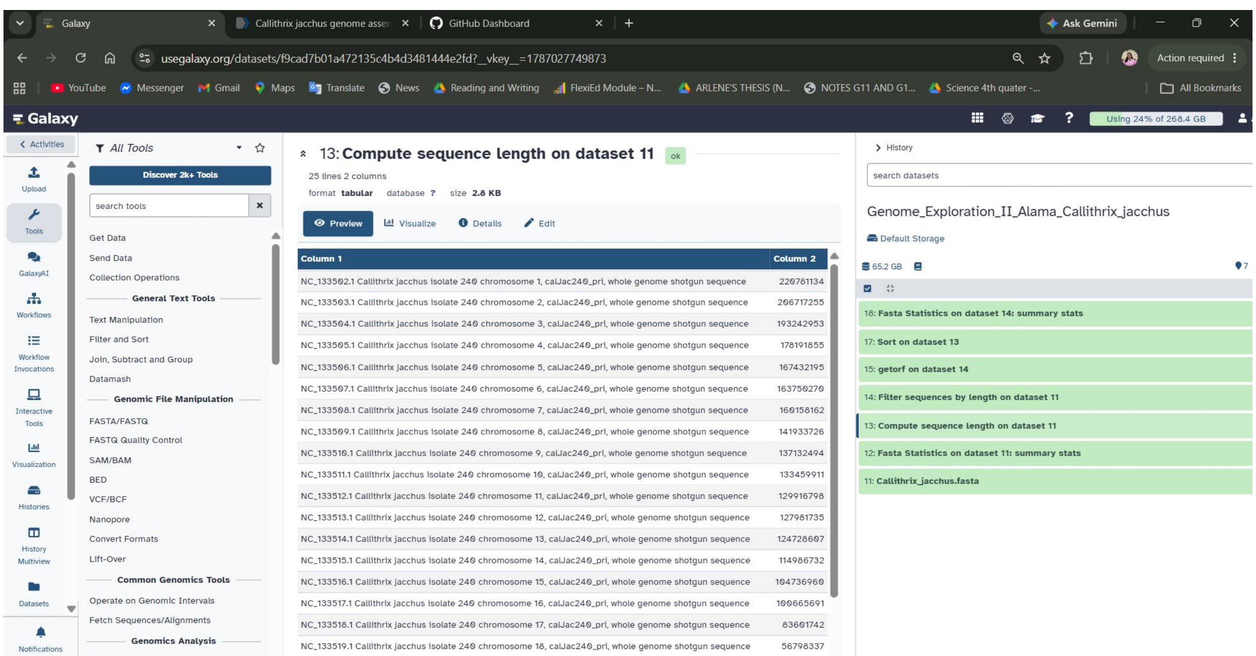


**Figure 3.** This shows the result of compute sequence length.

**Part 4. Length-Filtering Experiment (≥10 kb)**

| Metric | Original genome | After ≥10 kb filter |
|---|---|---|
| Total length | 2,933,425,964 bp (~2.9GB) | 2,933,425,964 bp (~2.9GB) |
| Number of sequences | 2,933,425,964 bp (~2.9GB) | 2,933,425,964 bp (~2.9GB) |
| Maximum length | 25 | 25 |
| N50 | 220,781,134 bp | 220,781,134 bp |
| GC content | 141,933,726 | 141,933,726 |

- **How many sequences were removed by the 10 kb filter?**

  - None. No sequences were removed, since the smallest sequence in the assembly (16,499 bp) already exceeds the 10 kb threshold.

- **Did the total assembly length decrease greatly or only slightly?**

  - It didn't decrease at all. The total length stayed exactly the same, because no sequences fell below the filter cutoff.

- **Did N50 change? Why do you think it changed or stayed similar?**

  - N50 stayed exactly the same (141,933,726 bp). This makes sense because N50 is driven by the longest sequences that together make up half the genome and filtering out very short sequences which this assembly doesn't have below 10 kb, has no effect on the large scaffolds that determine N50.

- **What does this tell you about the contribution of short sequences to your assembly?**

  - That this assembly has no short fragments below 10 kb and the genome is built almost entirely from substantial, chromosome-scale sequences.

**Part 5. Small ORF Exploration**

| Field | Value |
|---|---|
| **Sequence/region selected** | First ~19.6 kb of NC_133502.1 (Callithrix jacchus isolate 240, chromosome 1) |
| **Tool used** | EMBOSS getorf |
| **Minimum ORF size used** | ~300 bp |
| **Number of ORFs found** | 402 |
| **Length of longest ORF** | 1,191 bp |

- **How many ORFs were found in the selected sequence/region?**

  - 402 ORFs were found in the selected ~19.6 kb region of chromosome 1.

- **What was the longest ORF?**

  - The longest ORF was NC_133502.1_14, spanning positions 1773–2963, giving a length of 1,191 bp.

- **Why should you NOT report every ORF as a real gene?**

  -An ORF is simply a stretch of sequence readable without an internal stop codon under the chosen criteria; confirming a real gene requires additional evidence such as annotation, transcript data, protein similarity, or gene-prediction methods.

## Part 6. Short Biological Interpretation

This assembly of *Callithrix jacchus* is highly contiguous rather than fragmented, based on the sequence-length structure observed.

This assembly's near-complete, chromosome-scale contiguity means researchers can reliably study gene order, regulatory regions, and chromosome organization in Callithrix jacchus without worrying about artificial breaks distorting the biology. The near-absence of short fragments means little noise to complicate annotation or comparative genomics, and the typical mammalian GC content suggests a clean, uncontaminated genome.

The ORF exercise highlights a key limitation of genomics in which sequence pattern alone can flag a possible protein-coding region, but can't confirm a real gene that requires further evidence like expression data or homology to known proteins.